$$
\begin{aligned}
\nu(\tau) &= \frac{1}{2}\sum_i c_i^2 \left\langle \{\hat{q}_i(\tau), \hat{q}_i\} \right\rangle_{\hat{\rho}\mathcal{E}} \\
&= \sum_i \frac{c_i^2}{2m_i\omega_i} \coth\left(\frac{\omega_i}{2k_{\mathrm{B}}T}\right) \cos\left(\omega_i\tau\right) \\
&\equiv \int_0^\infty \mathrm{d}\omega\, J(\omega) \coth\left(\frac{\omega}{2k_{\mathrm{B}}T}\right) \cos\left(\omega\tau\right), \qquad (5.16) \\
\eta(\tau) &= \frac{\mathrm{i}}{2}\sum_i c_i^2 \left\langle [\hat{q}_i(\tau), \hat{q}_i] \right\rangle_{\hat{\rho}\mathcal{E}} \\
&= \sum_i c_i^2 \frac{1}{2m_i\omega_i} \sin\left(\omega_i\tau\right) \\
&\equiv \int_0^\infty \mathrm{d}\omega\, J(\omega) \sin\left(\omega\tau\right) \qquad (5.17)
\end{aligned}
$$